

NPDES PERMIT REVIEW SUMMARY

Facility:	Cadillac WWTP
Permit Number:	MI0020257
Reviewer:	Glen Schmitt, Permits Section, WRD, EGLE
Date:	9/30/2025

Outf.	Authorized Flow		Disch. Type *	Receiving Water	Lake?	FLOWS (cfs)			pH	Hardness
	mgd	cfs				95% Exc.	H. Mean	90Q10		
001	3.2	4.95		Clam River	☐	2.6	13.0	3.4	7	70

* P = Process; N = Noncontact Cooling; C = Contact Cooling; S = Sanitary; ST = Storm Water; GW = Groundwater Purge

SOURCES REVIEWED

TYPE OF REVIEW: IPP Local Limits Review

LOW FLOW FILE NO.: 10194 (Wexford County)

HARDNESS SOURCE: Collected from the Clam River on 8/19/2021 by Permits Staff at Latitude 44.236676, Longitude -85.4011 (STORET 830054).

PEL SHEET VERSION: 2-5-2025

Review the Following Information Sources

☐ **Permit Application**

☐ **Facility/VGW File**

☐ **MiEnviro Portal Documents & Files**

☐ **Current Permit**

☐ **WTA's**

☐ **Staff Report #'s**

☐ **WET Tests**

☐ **Other**

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 1 of 2)

*concentration values in ug/L except as noted

Facility:	Cadillac WWTP	Hardness:	70 mg/L CaCO3
Outfall:	001	pH:	7
Date:	9/30/2025	Drinking Water:	n

Parameter		CAS #	Water Quality Values										Diss. Met. Translator	Background Conc.
			FCV	vd	HNH	vd	HCV	vd	WV	vd	FAV	vd		
Beryllium		7440417	3.6566270011	2201401	1200	1201411	NA		NA		65.8145631353	2201401	1	0
Benzidine #		92875	2.7	2200902	3700	1201003	0.073	1201003	NA		49	2200902	1	
Bromomethane		74839	4.2	2201401	2600	1199708	NA		NA		75	2201401	1	
Phenol		108952	450	2200303	1200	2200310	NA		NA		6800	1200303	1	
	#N/A	1	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
	#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A

= Carcinogen

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 2 of 2)

**concentration values in ug/L except as noted; loads in lb/d*

Facility:	Cadillac WWTP	Disch. Rate:	4.95 cfs	95% Ex. Flow:	2.6 cfs	Conc.	Load (lbs/day)
Outfall:	001			H. Mean Flow:	13 cfs	<i>Hg Loading Calculated based on LCA (ng/L) of :</i>	0
Date:	9/30/2025			90Q10 Flow:	3.4 cfs	<i>TP Loading based on concentration (mg/L) of :</i>	0

Parameter	Monthly Average PEL								Daily Max PEL		PEQ		DECISION	
	FCV	load	HNV	load	HCV	load	WV	load	conc	load	Avg	Max	Avg	Max
Beryllium	4.1364450254	0.1103934	1987.3125	53.037396	#VALUE!	#VALUE!	#VALUE!	#VALUE!	65.814563	1.7564591	0	0		
Benzidine #	3.054290625	0.0815129	6127.546875	163.53197	0.1208948438	0.0032264416	#VALUE!	#VALUE!	49	1.307712	0	0		
Bromomethane	4.75111875	0.1267979	4305.84375	114.91436	#VALUE!	#VALUE!	#VALUE!	#VALUE!	75	2.0016	0	0		
Phenol	509.0484375	13.585485	1987.3125	53.037396	#VALUE!	#VALUE!	#VALUE!	#VALUE!	6800	181.4784	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		

* SILVER - Appropriate Dissolved Metal Translator must be entered in PEL 1

PEQ -- Effluent Data Summary
(all values in ug/L)

DATA ENTRY

[illegible]

PEQ -- Effluent Data Summary
(all values in ug/L)

[illegible]

PEQ -- Effluent Data Summary
(all values in ug/L)

Parameter:		Source	#N/A	Source	#N/A	Source	#N/A	Source
No. Nondetect			0		0		0	
Data								

PROJECTED EFFLUENT QUALITY ANALYSIS (<10 Detectable):

* all values in ug/l

	Beryllium	Benzidine #	Bromomethane	Phenol	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
Sample Max.	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total Samples	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Multiplier	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
PEQ:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PROJECTED EFFLUENT QUALITY ANALYSIS (>=10 Detectable):

PARAMETER:	Beryllium		Benzidine #		Bromomethane		Phenol		#N/A		#N/A		#N/A
TOTAL NO. VALUES:	0		0		0		0		0		0		0
DETECTED:	0		0		0		0		0		0		0
NON-DETECTED:	0		0		0		0		0		0		0
d (% data<detect)	#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!
m (mean of data >detect)	#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!
std dev	#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!
n	1	30	1	30	1	30	1	30	1	30	1	30	1
d^n	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
p	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
std normal dist fn (Z)	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
Z_p	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
1+(s/m)^2	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
(sigma_d)^2	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
mu_d	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
(sigma_dn)^2	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
mu_dn	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
P_95 exponent	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
P_95 (PEQ)	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max

NOTES: For purposes of this summary, ^ represents an exponent or superscript while _ represents a subscript.
Variables are defined in Part 12, Rule 1225 (3) (a).

[illegible]

[illegible]

CAS #	PARAMETER NAME
4860031	1-Chlorohexadecane
90120	1-Methylnaphthalene #
75343	1,1-Dichloroethane
75354	1,1-Dichloroethylene
71556	1,1,1-Trichloroethane
630206	1,1,1,2-Tetrachloroethane #
79005	1,1,2-Trichloroethane #
76131	1,1,2-Trichloro-1,2,2-trifluoroethane
79345	1,1,2,2-Tetrachloroethane #
632224	1,1,3,3-Tetramethylurea
112265	1,2-Bis(2-chloroethoxy)ethane
96128	1,2-Dibromo-3-chloropropane #
106934	1,2-Dibromoethane #
95501	1,2-Dichlorobenzene
107062	1,2-Dichloroethane #
540590	1,2-Dichloroethylene
156592	<i>cis</i> -1,2-Dichloroethylene
156605	<i>trans</i> -1,2-Dichloroethylene
78875	1,2-Dichloropropane #
542756	1,3-Dichloropropylene #
10061015	<i>cis</i> -1,3-Dichloropropylene #
10061026	<i>trans</i> -1,3-Dichloropropylene #
87616	1,2,3-Trichlorobenzene
526738	1,2,3-Trimethylbenzene
120821	1,2,4-Trichlorobenzene
95636	1,2,4-Trimethylbenzene
634662	1,2,3,4-Tetrachlorobenzene @
95943	1,2,4,5-Tetrachlorobenzene @
541731	1,3-Dichlorobenzene
108678	1,3,5-Trimethylbenzene
106467	1,4-Dichlorobenzene #
123911	1,4-Dioxane #
575371	1,7-Dimethylnaphthalene
95512	2-Chloroaniline
95578	2-Chlorophenol
104767	2-Ethylhexanol #
611143	2-Ethyltoluene
591786	2-Hexanone
88697	2-Isopropylphenol
91576	2-Methylnaphthalene
95487	2-Methylphenol
88755	2-Nitrophenol
67630	2-Propanol
594207	2,2-Dichloropropane
77769	2,2-Dimethoxypropane
526750	2,3-Dimethylphenol
58902	2,3,4,6-Tetrachlorophenol
1746016	2,3,7,8-TCDD # @
94757	2,4-D

120832	2,4-Dichlorophenol
105679	2,4-Dimethylphenol
51285	2,4-Dinitrophenol
120956	2,4-Di- <i>tert</i> -pentylphenol
93765	2,4,5-T
88062	2,4,6-Trichlorophenol #
126863	2,4,7,9-Tetramethyl-5-decyne-4,7-diol
585342	3- <i>tert</i> -Butylphenol
620144	3-Ethyltoluene
618451	3-Isopropylphenol
108394	3-Methylphenol
91941	3,3'-Dichlorobenzidine #
95658	3,4-Dimethylphenol
40360449	3,5,6-Trichloropicolinic acid
1702176	3,6-Dichloropicolinic acid
59507	4-Chloro-3-methylphenol
106478	4-Chloroaniline #
98668	4-Chlorobenzene sulfonic acid
106489	4-Chlorophenol
99898	4-Isopropylphenol
108112	4-Methyl-2-pentanol
108101	4-Methyl-2-pentanone
128370	4-Methyl-2,6-di-T-butylphenol #
106445	4-Methylphenol
100027	4-Nitrophenol
84852153	4-Nonylphenol
4684940	6-Chloropicolinic acid
208968	Acenaphthylene
83329	Acenaphthene
75070	Acetaldehyde
71501	Acetate (includes acetic acid)
64197	Acetic acid (includes acetate)
67641	Acetone
75058	Acetonitrile
98862	Acetophenone
79061	Acrylamide #
107131	Acrylonitrile #
15972608	Alachlor #
309002	Aldrin #
7664417	Ammonia, total ammonia nitrogen
62533	Aniline #
120127	Anthracene
7440360	Antimony
7440382	Arsenic #
1912249	Atrazine
103333	Azobenzene #
495487	Azoxybenzene
7440393	Barium
71432	Benzene #
92875	Benzidine #

56553	Benzo(a)anthracene
50328	Benzo(a)pyrene #
205992	Benzo(b)fluoranthene #
191242	Benzo(g,h,i)perylene
7440417	Beryllium
92524	Biphenyl
80057	Bisphenol A
111444	Bis(2-chloroethyl)ether #
108601	Bis(2-chloroisopropyl)ether #
117817	Bis(2-ethylhexyl)phthalate #
7440428	Boron
15541454	Bromate #
7726956	Bromine
74975	Bromochloromethane
75274	Bromodichloromethane #
75252	Bromoform #
74839	Bromomethane
71363	<i>n</i> -Butanol
85687	Butyl benzyl phthalate
109739	Butylamine
104518	<i>n</i> -Butylbenzene
135988	<i>sec</i> -Butylbenzene
98066	<i>tert</i> -Butylbenzene
7440439	Cadmium
63252	Carbaryl
86748	Carbazole #
75150	Carbon disulfide
56235	Carbon tetrachloride #
57749	Chlordane # @
16887006	Chloride
7782505	Chlorine
14998277	Chlorite
108907	Chlorobenzene
75003	Chloroethane #
67663	Chloroform #
74873	Chloromethane #
95498	<i>o</i> -Chlorotoluene
2921882	Chlorpyrifos
7440473	Chromium
18540299	Chromium, hexavalent
218019	Chrysene #
210880925	Clothianidin
7440484	Cobalt
7440508	Copper
21725462	Cyanazine #
57125	Cyanide, free
110827	Cyclohexane
221667318	Cyprosulfamide
10222012	DBNPA
50293	DDT # @

333415	Diazinon
53703	Dibenz(a,h)anthracene #
132649	Dibenzofuran
124481	Dibromochloromethane #
111922	Dibutylamine
75718	Dichlorodifluoromethane
75274	Dichlorobromomethane #
75434	Dichlorofluoromethane
62737	Dichlorvos
60571	Dieldrin # @
103231	Di(2-ethylhexyl)adipate #
84662	Diethyl phthalate
685916	Diethylacetamide
109897	Diethylamine
110816	Diethyldisulfide
111466	Diethylene glycol
124174	Diethylene glycol butyl ether acetate
108203	Diisopropylether
792745	Dimethyl 4,4'-biphenylcarboxylate
624920	Dimethyl disulfide
80466	Dimethylpropyl phenol
67685	Dimethylsulfoxide
84742	Di- <i>n</i> -butyl phthalate
117840	Di- <i>n</i> -octyl phthalate
84764	Dinonyl phthalate
88857	Dinoseb
101848	Diphenyloxide
2764729	Diquat
115297	Endosulfan
72208	Endrin
64175	Ethanol
637923	Ethyl <i>tert</i> -butyl ether
60297	Ethyl ether
75047	Ethylamine
100414	Ethylbenzene #
107211	Ethylene glycol
103361097	Flumioxazin
206440	Fluoranthene
86737	Fluorene
16984488	Fluoride
59756604	Fluridone
50000	Formaldehyde
64186	Formic acid
25812300	Gemfibrozil
76448	Heptachlor #
1024573	Heptachlor epoxide #
87821	Hexabromobenzene
118741	Hexachlorobenzene # @
87683	Hexachlorobutadiene # @
319846	<i>alpha</i> -Hexachlorocyclohexane # @

319857	<i>beta</i> -Hexachlorocyclohexane # @
319868	<i>delta</i> -Hexachlorocyclohexane @
77474	Hexachlorocyclopentadiene
67721	Hexachloroethane #
2691410	HMX
302012	Hydrazine #
7722841	Hydrogen peroxide
7783064	Hydrogen sulfide
15687271	Ibuprofen
138261413	Imidacloprid
288324	Imidazole
193395	Indeno (1,2,3-cd)pyrene #
64741668	Isopar C
78591	Isophorone #
98828	Isopropylbenzene
99876	<i>p</i> -Isopropyltoluene
7439921	Lead
58899	Lindane # @
7439932	Lithium
7439965	Manganese
950107	Mephosfolan
7439976	Mercury @
79390	Methacrylamide
67561	Methanol
40596698	Methoprene
74931	Methylmercaptan
625456	Methoxyacetic acid
108872	Methylcyclohexane
78933	Methyl ethyl ketone
1634044	Methyl <i>tert</i> butyl ether (MTBE) #
74895	Methylamine
75092	Methylene chloride #
51218452	Metolachlor #
2385855	Mirex # @
7439987	Molybdenum
38836394	N-(1-methylpropylidene)-2-propanamine
140807	N,N-diethyl-1,4-pentanediamine
927628	N,N-dimethyl-1-butamine
127195	N,N-dimethylacetamide
625503	N-ethyl acetamide
103695	N-ethylaniline
100618	N-methylaniline
91203	Naphthalene
22204531	Naproxen
7440020	Nickel
14797558	Nitrate, expressed as nitrogen
98953	Nitrobenzene #
25154523	<i>n</i> -Nonylphenol (mixed isomers)
872504	N-Methyl pyrrolidone
103651	N-Propylbenzene

706785	Octachlorocyclopentene
140669	Octylphenol
144627	Oxalic acid
10028156	Ozone
56382	Parathion
67774327	PBB #
1336363	PCB # @
608935	Pentachlorobenzene # @
87865	Pentachlorophenol #
79210	Peracetic acid
14797730	Perchlorate
375735	Perfluorobutanesulfonate
355464	Perfluorohexanesulfonate
375951	Perfluorononanonic acid
335671	Perfluorooctanoic acid
1763231	Perfluorooctane sulfonate @
85018	Phenanthrene
108952	Phenol
57410	Phenytoin #
32289580	PHMB
1918021	Picloram
51036	Piperonyl butoxide
7722647	Potassium permanganate
79094	Propionic acid
57556	Propylene glycol
75569	Propylene oxide #
129000	Pyrene
8003347	Pyrethrin
121824	RDX #
7782492	Selenium & inorganic salts
7440224	Silver
93721	Silvex
122349	Simazine
26628228	Sodium azide
140932	Sodium isopropyl xanthate
168316958	Spinosad
7440246	Strontium
100425	Styrene #
14808798	Sulfate
126863	Surfynol 104E
3383968	Temephos
51207319	Tetrachlorodibenzofuran, 2,3,7,8-
127184	Tetrachloroethylene #
109999	Tetrahydrofuran
7440280	Thallium
153719234	Thiamethoxam
7440326	Titanium
620235	m-Tolualdehyde
108883	Toluene
8001352	Toxaphene # @

102829	Tributylamine
79016	Trichloroethylene #
3380345	Triclosan
121448	Triethylamine
149735	Trimethylorthoformate
126727	Tris(2,3-dibromopropyl)phosphate #
7440622	Vanadium
75014	Vinyl chloride #
1330207	Xylene
7440666	Zinc

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CAS #	PARAMETER NAME	#	%	Background Conc.	HNV Drink	HNV Non-drink	WV	HCV Drink	HCV Non-drink	FCV		AMV		FAV	
					Value	Value	Value	Value	Value	Value	date	Value	date	Value	date
50000	Formaldehyde				5000 ###	###	NA	NA	NA	180	###	790	###	1600	###
50293	DDT # @		1		0 0.002 ###	0.002 ###	1E-005 ###	0.00015 ###	0.0002 ###		0.0032 ###		0.029 ###		0.057 ###
50328	Benzo(a)pyrene #				NLS	NLS	NA	NLS	NLS		2.7 ###		25 ###		49 ###
51036	Piperonyl butoxide				NLS	NLS	NA	NA	NA		0.45 ###		4.0 ###		8.1 ###
51285	2,4-Dinitrophenol			1	0 55 ###	2800 ###	NA	NA	NA		19 ###		130 ###		270 ###
53703	Dibenz(a,h)anthracene #				NLS	NLS	NA	NLS	NLS	ID	###	ID	###	ID	###
56235	Carbon tetrachloride #		1	1	0 18 ###	140 ###	NA	4.7 ###	38 ###		77 ###		690 ###		1400 ###
56382	Parathion				0 17 ###	54 ###	NA	NA	NA	0.013^	###	0.065^	###	0.13^	###
56553	Benzo(a)anthracene				NLS	NLS	NA	NLS	NLS		0.43 ###		3.9 ###		7.8 ###
57125	Cyanide, free			1	0 600 ###	48000 ###	NA	NA	NA		5.2 ###		22 ###		44 ###
57410	Phenytoin #				790 ###	11000 ###	NA	6.1 ###	89 ###		120 ###		1100 ###		2200 ###
57556	Propylene glycol				###	###	NA	NA	NA		2900000 ###		1000000 ###		2100000 ###
57749	Chlordane # @		1		0 0.0014 ###	0.0014 ###	NLS	0.00025 ###	0.0003 ###		0.029 ###		0.27 ###		0.53 ###
58899	Lindane # @		1	1	0 0.47 ###	0.5 ###	0.026 ###	0.025 ###	0.027 ###		0.07 ###		0.95 ###		1.9 ###
58902	2,3,4,6-Tetrachlorophenol				120 ###	150 ###	NA	NA	NA		1.2 ###		11 ###		22 ###
59507	4-Chloro-3-methylphenol				6900 ###	39000 ###	NA	NA	NA		7.4 ###		67 ###		130 ###
60297	Ethyl ether				14000 ###	###	NA	NA	NA		830 ###		7500 ###		15000 ###
60571	Dieldrin # @		1	1	0 0.0004 ###	0.0004 ###	7E-005 ###	7E-006 ###	7E-006 ###		0.056 ###		0.24 ###		0.48 ###
62533	Aniline #				190 ###	13000 ###	NA	21 ###	1500 ###		3 ###		10 ###		21 ###
62737	Dichlorvos				NLS	NLS	NA	NA	NA		0.0040 ###		0.038 ###		0.076 ###
63252	Carbaryl				NLS	NLS	NA	NA	NA		0.5200 ###		2.3 ###		4.5 ###
64175	Ethanol				###	###	NA	NA	NA	NLS		NLS		NLS	
64186	Formic acid				38000 ###	###	NA	NA	NA		14000 ###		120000 ###		240000 ###
64197	Acetic acid (includes Acetate)				16000 ###	###	NA	NA	NA		7697.1083627131 ###		69265.345834637 ###		138538.008976086 ###
67561	Methanol				14000 ###	###	NA	NA	NA		590000 ###		1300000 ###		2700000 ###
67630	2-Propanol				28000 ###	###	NA	NA	NA		57000 ###		500000 ###		1000000 ###
67641	Acetone				5600 ###	###	NA	NA	NA		1700 ###		15000 ###		30000 ###
67663	Chloroform #		1	1	0 350 ###	11000 ###	NA	* ###	* ###		630 ###		5700 ###		11000 ###
67685	Dimethylsulfoxide				###	###	NA	NA	NA		1900000 ###		1700000 ###		3400000 ###
67721	Hexachloroethane #		1		0 6 ###	7.6 ###	NA	5.3 ###	6.7 ###		13 ###		110 ###		210 ###
71363	n-Butanol				3500 ###	###	NA	NA	NA		9800 ###		88000 ###		180000 ###
71432	Benzene #		1	1	0 19 ###	510 ###	NA	12 ###	310 ###		200 ###		950 ###		1900 ###
71501	Acetate (includes Acetic Acid)				16000 ###	###	NA	NA	NA		7697.1083627131 ###		69265.345834637 ###		138538.008976086 ###
71556	1,1,1-Trichloroethane			1	0 62000 ###	###	NA	NA	NA		89 ###		800 ###		1600 ###
72208	Endrin				0 NLS	NLS	NA	NA	NA		0.036 ###		0.086 ###		0.172 ###
74839	Bromomethane				39 ###	2600 ###	NA	NA	NA		4.2 ###		38 ###		75 ###
74873	Chloromethane #				3500 ###	###	NA	110 ###	7300 ###		490 ###		4400 ###		8900 ###
74895	Methylamine				31 ###	2500 ###	NA	NA	NA		41 ###		370 ###		740 ###
74931	Methylmercaptan				140 ###	11000 ###	NA	NA	NA		1200 ###		11000 ###		22000 ###
74975	Bromochloromethane				1000 ###	59000 ###	NA	NA	NA		610 ###		5500 ###		11000 ###
75003	Chloroethane #		1		0 ###	###	NA	170 ###	9400 ###		1100 ###		10000 ###		20000 ###
75014	Vinyl chloride #		1	1	0 83 ###	4400 ###	NA	0.25 ###	13 ###		930 ###		8400 ###		17000 ###
75047	Ethylamine				740 ###	60000 ###	NA	NA	NA		31 ###		300 ###		610 ###
75058	Acetonitrile				5600 ###	###	NA	NA	NA		13000 ###		120000 ###		230000 ###
75070	Acetaldehyde				1200 ###	93000 ###	NA	NA	NA		130 ###		1200 ###		2400 ###
75092	Methylene chloride #		1	1	0 1600 ###	90000 ###	NA	47 ###	2600 ###		1500 ###		8500 ###		17000 ###
75150	Carbon disulfide				1000 ###	34000 ###	NA	NA	NA		9.0 ###		81 ###		160 ###
75252	Bromoform #				470 ###	8100 ###	NA	52 ###	890 ###		160 ###		1400 ###		2800 ###
75274	Bromodichloromethane #				170 ###	4500 ###	NA	6.8 ###	180 ###		180 ###		1700 ###		3300 ###
75343	1,1-Dichloroethane				0 9800 ###	###	NA	NA	NA		740 ###		6600 ###		13000 ###
75354	1,1-Dichloroethylene		1	1	0 1200 ###	33000 ###	NA	NA	NA		130 ###		1200 ###		2300 ###
75434	Dichlorodifluoromethane				2500 ###	###	NA	NA	NA		150 ###		1300 ###		2600 ###
75569	Propylene oxide #				ID*	ID*	NA	2.3 ###	180 ###		220 ###		2000 ###		4000 ###
75718	Dichlorodifluoromethane				2900 ###	90000 ###	NA	NA	NA		170 ###		1500 ###		3000 ###
76131	1,1,2-Trichloro-1,2,2-trifluoroethane				###	###	NA	NA	NA		32 ###		280 ###		570 ###
76448	Heptachlor #				0 ###	0.072 ###	NA	0.0017 ###	0.0018 ###		0.07 ###		0.42 ###		0.85 ###
77474	Hexachlorocyclopentadiene				140 ###	450 ###	NA	NA	NA		0.07 ###		0.27 ###		0.54 ###
77769	2,2-Dimethoxypropane				ID	ID	NA	NA	NA		6900 ###		62000 ###		120000 ###
78591	Isophorone #				4100 ###	###	NA	310 ###	8200 ###		1300 ###		4600 ###		9200 ###
78875	1,2-Dichloropropane #		1	1	0 12000 ###	###	NA	9.1 ###	290 ###		230 ###		2000 ###		4000 ###
78933	Methyl ethyl ketone				17000 ###	###	NA	NA	NA		2200 ###		20000 ###		40000 ###
79005	1,1,2-Trichloroethane		1	1	0 110 ###	3000 ###	NA	12 ###	330 ###		730 ###		3200 ###		6400 ###
79016	Trichloroethylene #		1	1	0 44 ###	550 ###	NA	29 ###	370 ###		200 ###		1800 ###		3500 ###
79061	Acrylamide #				28 ###	2200 ###	NA	0.12 ###	10 ###		590 ###		5300 ###		11000 ###
79094	Propionic acid				50000 ###	###	NA	NA	NA		3600 ###		33000 ###		65000 ###

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						Value date	Value date	Value date	Value date	Value date			
79210		Peracetic acid				NLS	NLS	NA	NA	NA	50 ###	110 ###	230 ###
79345		1,1,2,2-Tetrachloroeth		1	1	0 180 ###	4300 ###	NA	3.2 ###	78 ###	200 ###	910 ###	1800 ###
79390		Methacrylamide				NLS	NLS	NLS	NLS	NLS	4300 ###	38000 ###	77000 ###
80057		Bisphenol A				NLS	NLS	NA	NA	NA	20 ###	180 ###	350 ###
80466		Dimethylpropyl phenol				26 ###	38 ###	NA	NA	NA	5.8 ###	52 ###	100 ###
83329		Acenaphthene			1	0 580 ###	890 ###	NA	NA	NA	38 ###	100 ###	200 ###
84662		Diethyl phthalate				14000 ###	40000 ###	NA	NA	NA	110 ###	980 ###	2000 ###

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84742		Di-n-butyl phthalate				640 ###	690 ###	NA	NA	NA	9.7 ###	38 ###	75 ###
84764		Dinonyl phthalate				ID* ###	ID* ###	NA	NA	NA	140 ###	1200 ###	2500 ###
85018		Phenanthrene				0 ID* ###	ID* ###	NA	NA	NA	1.7 ###	5.7 ###	11 ###
85687		Butyl benzyl phthalate				6.9 ###	160 ###	NA	NA	NA	67 ###	310 ###	630 ###
86737		Fluorene				140 ###	160 ###	NA	NA	NA	12 ###	110 ###	220 ###
86748		Carbazole #				ID* ###	ID* ###	NA	19 ###	41 ###	4 ###	36 ###	72 ###
87616		1,2,3-Trichlorobenzene				55 ###	73 ###	NA	NA	NA	4.5 ###	40 ###	81 ###
87683		Hexachlorobutadiene		1		0 0.093 ###	0.098 ###	0.053 ###	0.33 ###	0.35 ###	1 ###	7 ###	15 ###
87821		Hexabromobenzene				81 ###	6500 ###	NA	NA	NA	ID ###	ID ###	ID ###
87865		Pentachlorophenol #		1	1	0 300 ###	450 ###	NA	1.8 ###	2.8 ###	6.6925836808 ###	8.7233208775 ###	17.446641755 ###
88062		2,4,6-Trichlorophenol		1	1	0 1900 ###	14000 ###	NA	41 ###	290 ###	5 ###	39 ###	79 ###
88697		2-Isopropylphenol				23 ###	380 ###	NA	NA	NA	36 ###	320 ###	650 ###
88755		2-Nitrophenol				78 ###	3100 ###	NA	NA	NA	56 ###	510 ###	1000 ###
88857		Dinoseb				28 ###	1900 ###	NA	NA	NA	0.48 ###	4.8 ###	9.5 ###
90120		1-Methylnaphthalene #				740 ###	1200 ###	NA	34 ###	54 ###	9.5 ###	86 ###	170 ###
91203		Naphthalene				0 1100 ###	2300 ###	NA	NA	NA	11 ###	100 ###	200 ###
91576		2-Methylnaphthalene				600 ###	1000 ###	NA	NA	NA	19 ###	170 ###	340 ###
91941		3,3'-Dichlorobenzidine		1		650 ###	950 ###	NA	0.14 ###	0.2 ###	4.5 ###	41 ###	81 ###
92524		Biphenyl				460 ###	690 ###	NA	NA	NA	13 ###	54 ###	110 ###
92875		Benzidine #				74 ###	3700 ###	NA	0.0015 ###	0.073 ###	2.7 ###	25 ###	49 ###
93721		Silvex				83 ###	140 ###	NA	NA	NA	30 ###	270 ###	540 ###
93765		2,4,5-T				490 ###	1200 ###	NA	NA	NA	14 ###	130 ###	260 ###
94757		2,4-D				430 ###	4700 ###	NA	NA	NA	270 ###	1700 ###	3500 ###
95487		2-Methylphenol				1400 ###	44000 ###	NA	NA	NA	76 ###	690 ###	1400 ###
95498		o-Chlorotoluene				360 ###	970 ###	NA	NA	NA	9.8 ###	88 ###	180 ###
95501		1,2-Dichlorobenzene			1	0 2000 ###	11000 ###	NA	NA	NA	13 ###	120 ###	240 ###
95512		2-Chloroaniline				91 ###	3100 ###	NA	NA	NA	24 ###	210 ###	430 ###
95578		2-Chlorophenol			1	0 190 ###	400 ###	NA	NA	NA	18 ###	160 ###	320 ###
95636		1,2,4-Trimethylbenzene				190 ###	330 ###	NA	NA	NA	17 ###	150 ###	310 ###
95658		3,4-Dimethylphenol				37 ###	580 ###	NA	NA	NA	25 ###	230 ###	460 ###
95943		1,2,4,5 - Tetrachlorobenzene @				2.8 ###	2.9 ###	NA	NA	NA	3 ###	23 ###	46 ###
96128		1,2-Dibromo-3-chloropropane #				ID* ###	ID* ###	NA	0.24 ###	4.9 ###	75 ###	670 ###	1300 ###
98066		tert-Butylbenzene				82 ###	110 ###	NA	NA	NA	ID ###	ID ###	ID ###
98668		4-Chlorobenzene sulfonic acid				28000 ###	###	NA	NA	NA	1300 ###	150000 ###	290000 ###
98828		Isopropylbenzene				1700 ###	3800 ###	NA	NA	NA	28 ###	250 ###	500 ###
98862		Acetophenone				5700 ###	###	NA	NA	NA	450 ###	4100 ###	8200 ###
98953		Nitrobenzene #				26 ###	990 ###	NA	4.7 ###	180 ###	230 ###	1000 ###	2100 ###
99876		p-Isopropyltoluene				33 ###	45 ###	NA	NA	NA	3.2 ###	29 ###	58 ###
99898		4-Isopropylphenol				21 ###	130 ###	NA	NA	NA	20 ###	180 ###	360 ###
###		4-Nitrophenol				680 ###	18000 ###	NA	NA	NA	200 ###	940 ###	1900 ###
###		Ethylbenzene #			1	0 1600 ###	6900 ###	NA	25 ###	110 ###	21 ###	190 ###	370 ###
###		Styrene #				4200 ###	18000 ###	NA	20 ###	80 ###	160 ###	1400 ###	2900 ###
###		N-Methylaniline				ID ###	ID ###	NA	NA	NA	360 ###	3200 ###	6,500 ###
###		Diphenyloxide				44 ###	78 ###	NA	NA	NA	2.9 ###	26 ###	52 ###
###		Tributylamine				0 ID ###	ID ###	NA	NA	NA	2.8 ###	25 ###	50 ###
###		Di(2-ethylhexyl)adipate #				3500 ###	6500 ###	NA	150 ###	280 ###	4.6 ###	41 ###	83 ###
###		Azobenzene #				ID* ###	ID* ###	NA	3.7 ###	6.0 ###	4.1 ###	37 ###	74 ###
###		N-Propylbenzene				150 ###	290 ###	NA	NA	NA	4.9 ###	44 ###	89 ###
###		N-Ethylaniline				7.8 ###	190 ###	NA	NA	NA	1.8 ###	16 ###	32 ###
###		n-Butylbenzene				48 ###	57 ###	NA	NA	NA	2.7 ###	24 ###	48 ###
###		2-Ethylhexanol #				8700 ###	68000 ###	NA	86 ###	660 ###	130 ###	1100 ###	2300 ###
###		2,4-Dimethylphenol			1	0 450 ###	8700 ###	NA	NA	NA	380 ###	1300 ###	2700 ###
###		4-Methylphenol				1400 ###	45000 ###	NA	NA	NA	25 ###	230 ###	450 ###
###		1,4-Dichlorobenzene #		1	1	0 1100 ###	11000 ###	NA	24 ###	240 ###	17 ###	100 ###	210 ###
###		4-Chloroaniline #				39 ###	1400 ###	NA	2 ###	72 ###	19 ###	170 ###	340 ###
###		4-Chlorophenol				880 ###	14000 ###	NA	NA	NA	30 ###	270 ###	530 ###
###		1,2-Dibromoethane #				250 ###	8200 ###	NA	0.17 ###	5.7 ###	15 ###	140 ###	280 ###
###		1,2-Dichloroethane #		1	1	0 6900 ###	###	NA	6.0 ###	360 ###	2000 ###	8200 ###	16000 ###
###		Acrylonitrile #		1	1	0 58 ###	320 ###	NA	0.21 ###	1.2 ###	66 ###	590 ###	1200 ###
###		Ethylene glycol				56000 ###	###	NA	NA	NA	190000 ###	1700000 ###	3400000 ###
###		4-Methyl-2-pentanone				1400 ###	85000 ###	NA	NA	NA	1000 ###	6700 ###	13000 ###
###		4-Methyl-2-pentanol				1400 ###	85000 ###	NA	NA	NA	320 ###	2800 ###	5900 ###
###		Diisopropylether				7300 ###	###	NA	NA	NA	470 ###	4200 ###	8400 ###
###		3-Methylphenol				2700 ###	89000 ###	NA	NA	NA	71 ###	636 ###	1271 ###
###		Bis(2-chloroisopropyl)ether #				990 ###	47000 ###	NA	6 ###	290 ###	540 ###	4900 ###	9700 ###

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						Value	verif date	Value	verif date	Value	verif date	Value	verif date	Value	verif date	Value	verif date	Value	verif date	Value	verif date
###	1,3,5-Trimethylbenzene					2000	###	4200	###	NA		NA		NA		45	###	410	###	810	###
###	Methylcyclohexane					34	###	54	###	NA		NA		NA		3.1	###	28	###	55	###
###	Toluene				1	0	5600	###	51000	###	NA	NA		NA		260	###	1300	###	2600	###
###	Chlorobenzene					0	470	###	3200	###	NA	NA		NA		25	###	220	###	450	###
###	Phenol						1100	###	1200	###	NA	NA		NA		450	###	3,400	###	6,800	###
###	Butylamine						45	###	3100	###	NA	NA		NA		57	###	510	###	1000	###

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					Value	Value	Value	Value	Value			
###	Diethylamine				310 ###	24000 ###	NA	NA	NA	20 ###	180 ###	350 ###
###	Tetrahydrofuran				350 ###	26000 ###	NA	NA	NA	11000 ###	74000 ###	150000 ###
###	Diethyldisulfide				50 ###	380 ###	NA	NA	NA	62 ###	560 ###	1100 ###
###	Cyclohexane				84000 ###	###	NA	NA	NA	15 ###	140 ###	280 ###
###	Bis(2-chloroethyl)ether #				ID* ###	ID* ###	NA	0.79 ###	15 ###	ID*	9200 ###	18000 ###
###	Diethylene glycol				###	###	NA	NA	NA	43000 ###	380000 ###	770000 ###
###	Dibutylamine				6.7 ###	69 ###	NA	NA	NA	36 ###	320 ###	650 ###
###	1,2-Bis(2-chloroethoxy)ethane				6.9 ###	460 ###	NA	NA	NA	1500 ###	13000 ###	26000 ###
###	Endosulfan				85 ###	170 ###	NA	NA	NA	0.029 ###	0.13 ###	0.27 ###
###	Bis(2-ethylhexyl)phthalate #				120 ###	160 ###	NA	14 ###	18 ###	ID*	285 ###	285 ###
###	Di-n-octyl phthalate				240 ###	300 ###	NA	NA	NA	ID	ID	ID
###	Hexachlorobenzene #		1	1	0 0.046 ###	0.046 ###	0.0003 ###	0.00045 ###	0.0005 ###	ID*	610 ###	1200 ###
###	Anthracene				1900 ###	2400 ###	NA	NA	NA	3.2 ###	29 ###	58 ###
###	1,2,4-Trichlorobenzene				80 ###	99 ###	NA	NA	NA	130 ###	420 ###	850 ###
###	2,4-Dichlorophenol			1	0 220 ###	1100 ###	NA	NA	NA	11 ###	92 ###	180 ###
###	2,4-Di-tert-pentylphenol				0.58 ###	0.61 ###	NA	NA	NA	NLS	NLS	NLS
###	Triethylamine				4000 ###	###	NA	NA	NA	260 ###	1100 ###	2100 ###
###	RDX #				83 ###	6100 ###	NA	5.8 ###	420 ###	85 ###	400 ###	790 ###
###	Simazine				140 ###	4000 ###	NA	NA	NA	17 ###	160 ###	310 ###
###	1,4-Dioxane #				890 ###	72000 ###	NA	3.5 ###	280 ###	22000 ###	200000 ###	390000 ###
###	Diethylene glycol butyl ether acetate				330 ###	13000 ###	NA	NA	NA	260 ###	2300 ###	4600 ###
###	Dibromochloromethane #				570 ###	12000 ###	NA	6.8 ###	150 ###	170 ###	1600 ###	3100 ###
###	Tris(2,3-dibromopropyl)phosphate #				ID* ###	ID* ###	NA	0.31 ###	13 ###	1.4 ###	12 ###	25 ###
###	Tetramethyl decynediol				ID* ###	ID* ###	NA	NA	NA	350 ###	3,100 ###	6200 ###
###	Tetrachloroethylene #		1	1	0 60 ###	330 ###	NA	11 ###	60 ###	190 ###	1400 ###	2900 ###
###	N,N-Dimethylacetamide				700 ###	57000 ###	NA	NA	NA	4100 ###	37000 ###	74000 ###
###	4-Methyl-2,6-di-T-butylphenol #				3 ###	3 ###	NA	0.25 ###	0.25 ###	0.37 ###	3.3 ###	6.6 ###
###	Pyrene				15 ###	15 ###	NA	NA	NA	0.58 ###	5.2 ###	10 ###
###	Dibenzofuran				ID* ###	ID* ###	NA	NA	NA	4 ###	36 ###	72 ###
###	sec-Butylbenzene				77 ###	100 ###	NA	NA	NA	3.1 ###	28 ###	56 ###
###	Octylphenol				NLS	NLS	NA	NA	NA	2 ###	13 ###	26 ###
###	N,N-diethyl-1,4-pentanediamine				5 ###	360 ###	NA	NA	NA	2600 ###	23000 ###	46000 ###
###	Sodium isopropyl xanthate				22 ###	1800 ###	NA	NA	NA	150 ###	1400 ###	2800 ###
###	Oxalic acid				4500 ###	###	NA	NA	NA	250 ###	2300 ###	4600 ###
###	Trimethylorthoformate				87 ###	6700 ###	NA	NA	NA	1600 ###	15000 ###	29000 ###
###	cis-1,2-Dichloroethylene				880 ###	36000 ###	NA	NA	NA	150 ###	1400 ###	2800 ###
###	trans-1,2-Dichloroethylene		1		0 470 ###	19000 ###	NA	NA	NA	150 ###	1400 ###	2800 ###
###	Benzo(g,h,i)perylene				17 ###	19 ###	NA	NA	NA	3 ###	27 ###	54 ###
###	Indeno (1,2,3-cd)pyrene #				NLS	NLS	NA	NLS	NLS	ID	ID	ID
###	Benzo(b)fluoranthene #				NLS	NLS	NA	NLS	NLS	ID	ID	ID
###	Fluoranthene				18 ###	18 ###	NA	NA	NA	1.6 ###	14 ###	28 ###
###	Acenaphthylene				260 ###	530 ###	NA	NA	NA	6.6 ###	60 ###	120 ###
###	Chrysene #				ID* ###	ID* ###	NA	1.5 ###	1.5 ###	0.43 ###	3.9 ###	7.8 ###
###	Imidazole				27 ###	2200 ###	NA	NA	NA	1500 ###	13000 ###	26000 ###
###	Methyl Parathion				5 ###	88 ###	NA	NA	NA	0.013^	0.065^	0.13^
###	Hydrazine #				42 ###	3400 ###	NA	0.094 ###	7.6 ###	1.8 ###	16 ###	32 ###
###	Aldrin #				0.0001 ###	0.0001 ###	NA	9E-006 ###	9E-006 ###	0.017 ###	0.15 ###	0.3 ###
###	alpha-Hexachlorocyclohexane # @				19 ###	21 ###	ID* ###	0.016 ###	0.017 ###	ID	ID	ID
###	beta-Hexachlorocyclohexane # @				0.091 ###	0.098 ###	ID* ###	0.024 ###	0.026 ###	ID	ID	ID
###	delta-Hexachlorocyclohexane @				1 28 ###	30 ###	ID ###	NA	NA	ID	ID	ID
###	Diazinon				12 ###	37 ###	NA	NA	NA	0.004 ###	0.064 ###	0.13 ###
###	Perfluorooctanoic acid				0.066 ###	0.17 ###	NA	NA	NA	880 ###	7700 ###	15000 ###
###	Perfluorohexanesulfonate				0.059 ###	0.21 ###	NA	NA	NA	NLS	NLS	NLS
###	Perfluorobutanesulfonate				8.3 ###	670 ###	NA	NA	NA	24000 ###	120000 ###	240000 ###
###	Perfluorononanoic acid				0.019 ###	0.03 ###	NA	NA	NA	NLS	NLS	NLS
###	Azoxybenzene				17 ###	1400 ###	NA	NA	NA	ID	ID	ID
###	1,2,3-Trimethylbenzene				290 ###	650 ###	NA	NA	NA	6.5 ###	59 ###	120 ###
###	2,3-Dimethylphenol				16 ###	270 ###	NA	NA	NA	120 ###	1100 ###	2200 ###
###	1,2-Dichloroethylene				470 ###	19000 ###	NA	NA	NA	150 ###	1400 ###	2800 ###
###	1,3-Dichlorobenzene			1	0 37 ###	65 ###	NA	NA	NA	28 ###	100 ###	200 ###
###	1,3-Dichloropropylene				930 ###	39000 ###	NA	3.3 ###	140 ###	9 ###	81 ###	160 ###
###	1,7-Dimethylnaphthalene				ID ###	ID ###	NA	NA	NA	2.8 ###	26 ###	51 ###
###	3-tert-Butylphenol				110 ###	360 ###	NA	NA	NA	29 ###	260 ###	530 ###
###	2-Hexanone				9700 ###	###	NA	NA	NA	570 ###	5200 ###	10000 ###
###	2,2-Dichloropropane				ID ###	ID ###	NA	NA	NA	80 ###	720 ###	1400 ###

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Rule 57 Surface Water Quality Values
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Assessment, Restoration, and Management Section
Michigan EGLE
December 6, 2024

Verif Date	CAS #	PARAMETER NAME	#	%	Background Conc.	HNV Drink	HNV Non-drink	WV	HCV Drink	HCV Non-drink	FCV	AMV	FAV
						verif Value date	verif Value date	verif Value date	verif Value date	verif Value date			
###	###	Pentachlorobenzene # @				0.38 ###	0.38 ###	0.019 ###	NA	NA	1.2 ###	11 ###	21 ###
###	###	2-Ethyltoluene				71 ###	150 ###	NA	NA	NA	ID ###	ID ###	ID ###
###	###	3-Isopropylphenol				39 ###	250 ###	NA	NA	NA	26 ###	240 ###	470 ###
###	###	3-Ethyltoluene				67 ###	130 ###	NA	NA	NA	ID ###	ID ###	ID ###
###	###	m-Tolualdehyde				ID* ###	ID* ###	NA	NA	NA	34 ###	310 ###	620 ###
###	###	Dimethyl disulfide				0 480 ###	20000 ###	NA	NA	NA	ID ###	ID ###	ID ###
###	###	Methoxyacetic acid				28 ###	2200 ###	NA	NA	NA	270 ###	2400 ###	4800 ###
###	###	N-Ethyl acetamide				87 ###	7000 ###	NA	NA	NA	6900 ###	62000 ###	120000 ###
###	###	1,1,1,2-Tetrachloroethane #				680 ###	3500 ###	NA	19 ###	100 ###	85 ###	770 ###	1500 ###
###	###	1,1,3,3-Tetramethylurea				39 ###	3100 ###	NA	NA	NA	7700 ###	69000 ###	140000 ###
###	###	1,2,3,4-Tetrachlorobenzene @				3.9 ###	4.1 ###	ID ###	###	NA	3.4 ###	18 ###	35 ###
###	###	Ethyl <i>tert</i> -butyl ether				2500 ###	### ###	NA	NA	NA	350 ###	3100 ###	6300 ###
###	###	Diethylacetamide				44 ###	3400 ###	NA	NA	NA	6400 ###	58000 ###	120000 ###
###	###	Octachlorocyclopentene				0.014 ###	0.014 ###	NA	NA	NA	0.08 ###	0.75 ###	1.5 ###
###	###	Dimethyl 4,4'-biphenyl carboxylate				ID ###	ID ###	NA	NA	NA	ID ###	ID ###	ID ###
###	###	N-Methyl pyrrolidone				25000 ###	### ###	NA	NA	NA	6600 ###	59000 ###	120000 ###

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Rule 57 Surface Water Quality Values
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Assessment, Restoration, and Management Section
Michigan EGLE
December 6, 2024

Verif Date	CAS #	PARAMETER NAME	#	%	Background Conc.	HNV Drink verif date	HNV Non-drink verif date	WV verif date	HCV Drink verif date	HCV Non-drink verif date	FCV verif date	AMV verif date	FAV verif date
###	###	N,N-dimethyl-1-butamine				ID	###	ID	###	NA	510	###	9200
###	###	Mephosfolan				2.5	###	170	###	NA	0.37	###	6.6
###	###	Heptachlor epoxide #				0.0075	###	0.0076	###	0.0021	###	0.33	###
###	###	Xylene			1	0	3800	16000	###	NA	49	###	890
###	###	PCB # @		1	1 **	NLS	###	0.0001	###	3E-005	###	ID*	###
###	###	Methyl tert butyl ether (MTBE) #				920	###	63000	###	100	###	7100	###
###	###	3,6-Dichloropicolinic acid				4100	###	###	###	NA	32000	###	420000
###	###	2,3,7,8-TCDD # @		1		0	###	###	3E-009	###	660	###	12000
###	###	Perfluorooctane sulfonate @				0.011	###	0.012	###	NA	140	###	1600
###	###	Atrazine				880	###	8600	###	NA	7.3	###	100
###	###	Picloram				5500	###	###	###	NA	46	###	570
###	###	Mirex # @				*	###	*	###	2E-005	###	ID	###
###	###	HMX				1400	###	###	###	NA	250	###	4600
###	###	Diquat				61	###	4900	###	NA	6	###	110
###	###	Chlorpyrifos				*	*	*	###	NA	0.0020	###	0.053
###	###	Triclosan				NLS	NLS	NA	NA	NA	4.4000	###	50
###	###	Temephos				NLS	NLS	NA	NA	NA	0.0001	###	0.0013
###	###	6-Chloropicolinic acid				410	###	22000	###	NA	26	###	460
###	###	1-Chlorohexadecane				ID	###	ID	###	NA	ID	ID	ID
###	###	Lead			1	2.5	14	190	###	NA	18.5613506884	###	181.2870159605
###	###	Lithium			1	0	720	###	58000	###	440	###	1800
###	###	Manganese				1300	###	59000	###	NA	1437.0938887969	###	6201.0767208978
###	###	Mercury @		1 **		0.0018	###	0.0018	###	0.0013	###	1.4	###
###	###	Molybdenum		0		120	###	10000	###	NA	3200	###	58000
###	###	Nickel		1		3	2600	###	###	NA	38.4601432866	###	692.5442622761
###	###	Silver		1		0	130	###	11000	###	0.06	###	1.1
###	###	Strontium				ID*	###	ID*	###	NA	36000	###	81000
###	###	Thallium		1		0	1.2	###	3.7	###	7.2	###	94
###	###	Titanium				NLS	NLS	NA	NA	NA	ID	ID	ID
###	###	Antimony				1.7	###	130	###	NA	240	###	2300
###	###	Arsenic #		1	1	1	10	###	280	###	150	###	680
###	###	Barium		1		0	5800	###	###	NA	301.4263153525	###	1709.2767260084
###	###	Beryllium				0	160	###	1200	###	3.6566270011	###	65.8145631353
###	###	Boron				4000	###	###	###	NA	7200	###	69000
###	###	Cadmium		1		0	2.5	###	130	###	1.7190854143	###	5.7939008687
###	###	Chromium		1		1.6	120	###	9400	###	55.3399302243	###	850.8635018933
###	###	Cobalt				ID*	###	ID*	###	NA	100	###	740
###	###	Copper		1		4	470	###	38000	###	6.6029544302	###	19.2066659389
###	###	Vanadium				53	###	540	###	NA	27	###	160
###	###	Zinc		1		13	3300	###	16000	###	87.3262490635	###	173.2354393186
###	###	Ammonia, total ammonia nitrogen				ID*	###	ID*	###	NA	'see equation in foo	###	'see equation in footn
###	###	Potassium permanganate				NLS	NLS	NA	NA	NA	* ###	29	###
###	###	Hydrogen peroxide				ID*	###	ID*	###	NA	10	###	180
###	###	Bromine				ID*	###	ID*	###	NA	0.26	###	4.8
###	###	Selenium & inorganic salts		1		0	120	###	2700	###	5	###	120
###	###	Chlorine				NLS	NLS	NA	NA	NA	*	19	###
###	###	Hydrogen sulfide				160	###	13000	###	NA	0.36	###	6.4
###	###	Toxaphene # @		1	1	0	0.021	###	0.021	###	7E-005	###	0.3
###	###	Pyrethrin				NLS	NLS	NA	NA	NA	0.005	###	0.39
###	###	Ozone				NLS	NLS	NLS	NLS	NLS	0.022	###	1.2
###	###	trans-1,3-Dichloropropylene #				930	###	39000	###	NA	9.0	###	160
###	###	cis-1,3-Dichloropropylene #				930	###	39000	###	NA	9.0	###	160
###	###	DBNPA				ID*	###	ID*	###	NA	7.8	###	140
###	###	Nitrate, expressed as nitrogen				10000	###	NLS	NA	NA	NLS	NLS	NLS
###	###	Perchlorate				19	###	1600	###	NA	400	###	13000
###	###	Sulfate				ID*	###	ID*	###	NA	370000	###	1200000
###	###	Chlorite				830	###	67000	###	NA	0.72	###	13
###	###	Bromate #				100	###	8200	###	NA	760	###	14000
###	###	Ibuprofen				NLS	NLS	NA	NA	NA	96	###	1700
###	###	Alachlor #				270	###	6900	###	NA	11	###	300
###	###	Chloride				ID*	###	ID*	###	NA	150000	###	640000
###	###	Fluoride				NLS	NLS	NA	NA	NA	2113.7792857339	###	15478.8948888776
###	###	Chromium, hexavalent		1		0	120	###	9400	###	11	###	32
###	###	Cyanazine #				190	###	12000	###	NA	110	###	2000

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of both compounds						HNV Drink	HNV Non-drink	WV	HCV Drink	HCV Non-drink	FCV	AMV	FAV	
Verif Date	PARAMETER NAME	#	%	Background Conc.		verif Value date	verif Value date	Value date	verif Value date	verif Value date	Value date	verif Value date	verif Value date	
###	Naproxen					NLS	NLS	NA	NLS	NLS		96 ###	860 ###	1700 ###
###	n-Nonylphenol (mixed isomers)					NLS	NLS	NA	NA	NA		2 ###	7 ###	14 ###
###	Gemfibrozil					NLS	NLS	NA	NA	NA		44 ###	400 ###	800 ###
###	Sodium azide					330 ###	27000 ###	NA	NA	NA		7.3 ###	65 ###	130 ###
###	PHMB					10000 ###	###	NA	51 ###	4100 ###		0.2 ###	1.8 ###	3.6 ###
###	N-(1-methylpropylidene)-2-propanamine					ID ###	ID ###	NA	NA	NA		260 ###	2400 ###	4700 ###
###	3,5,6-Trichloropicolinic acid					ID ###	ID ###	NA	NA	NA	ID	###	ID	ID
###	Methoprene					21 ###	21 ###	NA	NA	NA		2.9 ###	26 ###	51 ###
###	Tetrachlorodibenzofuran, 2,3,7,8-			1		0 NLS	NLS	NA	NA	NA	ID	###	ID	ID
###	Metolachlor #					3300 ###	14000 ###	NA	78 ###	340 ###		15 ###	110 ###	210 ###
###	Fluridone					2200 ###	80000 ###	NA	NA	NA		1 ###	3 ###	250 ###
###	Isopar C					4500 ###	5600 ###	NA	NA	NA		1.9 ###	17 ###	35 ###
###	PBB #					0.0003 ###	0.0003 ###	NA	0.00013 ###	0.0001 ###	NLS	NLS	NLS	
###	4-Nonylphenol					130 ###	200 ###	NA	NA	NA	ID	###	ID	ID
###	Flumioxazin					NLS	NLS	NA	NA	NA		16 ###	140 ###	290 ###
###	Imidacloprid					NLS	NLS	NA	NLS	NLS		0.029 ###	0.36 ###	0.72 ###
###	Thiamethoxam					NLS	NLS	NA	NLS	NLS		0.28 ###	0.53 ###	1.1 ###
###	Spinosad					620 ###	7500 ###	NA	NA	NA		30 ###	250 ###	500 ###
###	Clothianidin					NLS	NLS	NA	NLS	NLS		0.081 ###	0.29 ###	0.58 ###
###	Cyprosulfamide					11000 ###	###	NA	NA	NA		680 ###	6300 ###	13000 ###

¹Ammonia Aquatic Life Equations (mg/L total ammonia nitrogen; TAN)
T = Temperature (in °C)
MAX = Maximum value in following parentheses

$4/(1+10^{(7.204-pH)})+1.6181/(1+10^{(pH-7.204)}))*(14.70*10^{(0.036*(20-MAX(T,13))}))$

$4/(1+10^{(7.204-pH)})+1.6181/(1+10^{(pH-7.204)}))*(14.70*10^{(0.036*(20-MAX(T,13))}))$

$/(1+10^{(7.688-pH)})+1.1994/(1+10^{(pH-7.688)}))*(2.276*10^{(0.028*(20-MAX(T,7))}))$